

Unit - I :

Introduction to Cognitive Science:

- understanding cognition.
- IBM's Watson.
- Design for human cognition.
- Augmented Intelligence.
- cognition modeling paradigms:

- Declarative / logic based computational cognitive modeling.

- Connectionist models of cognition.
- Bayesian models of cognition.
- A dynamical sys approach to cognition.

Unit - II

Cognitive Modeling:

- Modeling the interaction of language.
- Memory & Learning
- Modeling select aspects of cognition classical models of rationality.
- Symbolic reasoning & decision making.
- Application of cognitive computing & systems.

Unit - III

- cognitive models of memory & language.
- Computational models of episodic & semantic memory.
- Modeling psycholinguistics.

Unit - IV

- formal models of inductive generalizations.

- causality.

- categorization & similarity

- The role of analogy in problem solving.

- cognitive development child & concept acquisition.

- cognition & Artificial cognitive architectures such as ACT-R,

→ SOAR.

- open cog

- copy cat

- Memory N/w.

Unit - V

- Deep QA Architecture.

- Unstructured info management Architecture (UIMA).

- Structured knowledge.

- Business implications.

- Building cognitive applications.

language

Unit - I

Introduction to cognitive science + understanding cognition

→ 'Cognitive science is the study of how the human mind works, including thinking, Learning, memory & decision making.

- The goal is to understand how humans think & learn

- To study how the brain processes info.
- To develop sys that mimic human intelligence
- To improve interaction b/w humans & computers.

⇒ Cognitive science combines multiple fields:

• Psychology, CS, Neuroscience, Linguistics.

⇒ So, it is called an interdisciplinary subject

Understanding cognition :-

⇒ cognition refers to the mental processes involved in gaining knowledge & understanding.

⇒ cognition includes the following processes:

- perception → Receiving info. through senses (seeing, hearing, etc.)
- Attention → focusing on specific info.
- Memory → storing & recalling info.
- Learning → Acquiring new knowledge
- Thinking → processing info.
- Reasoning → making logical decisions.
- problem solving → finding solutions.
- language → understanding & communicating.

Type :-

① Conscious cognition :-

- Done with awareness.

ex: Solving a math problem.

② Unconscious cognition :-

- Happens automatically

ex: Recognizing a familiar face.

Cognitive Model:

- Human mind works like a computer:

Input $\rightarrow$ processing $\rightarrow$ output

IBM's Watson:

$\Rightarrow$ It is an AI system developed by IBM that can understand Natural language, process data & provide intelligent answers.

- It was introduced by IBM in 2011.
- It became famous after winning the quiz show Jeopardy!
- It competed against human champions & won.

Architecture:

- Watson works in the following steps:

(i) Input processing:

- Receives question in natural language.

(ii) Analysis:

- Breaks down the question.
- Identifies keywords.

(iii) Hypothesis Generation:

- Generates possible answers.

(iv) Evidence scoring:

- checks data & ranks answers.

(v) Final Answer:

- gives best possible answer.

Technologies used:

- AI
- ML
- NLP
- IRS
- Big Data Analytics.

Applications:

- Health care
- Banking
- Customer Service
- Education
- Business.

Advantages:

- Handles huge data quickly.
- provides accurate answers.
- Improves decision making.
- learns continuously.

Limitations:

- Expensive to Implement
- Needs large data.
- sometimes gives incorrect answers.

Design for Human cognition :

⇒ It means designing systems, interfaces or applications in a way that matches how humans think, learn, & process info.

— The Goal is to make sys easy to understand, use & interact with by aligning with human mental abilities.

→ Human cognition includes:

- Perception.
- Memory
- Attention
- Learning
- Decision making.

Design principles :

(i) Simplicity :

— Keep design simple & clear.

— Avoid unnecessary complexity.

ent Clean UI with fewer options.

(ii) Consistency :

— Similar elements should behave the same way.

— Helps users predict actions.

ent Same button style across app.

(iii) Visibility :

- Important elements should be easily visible.
- users should not struggle to find options.

ex: Highlight "submit" button.

(iv) Feedback :

- sys should give immediate response.
- Helps users to know what happened.

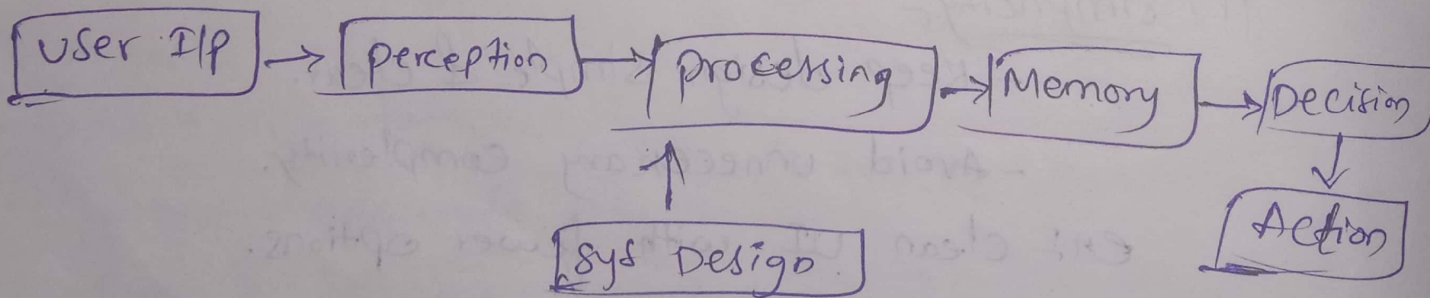
ex: "form submitted successfully" message

(v) Affordance :

- Design should suggest how to use it.

ex: Button looks clickable.

Diagram



Applications :

- user interfaces
- AI systems.
- website & mobile apps

- Advantages ↓
- Improves usability
 - Reduces errors
 - faster Learning.

Conclusion ↓

- It focuses on creating sys that match human thinking patterns, making interaction simple, efficient & user friendly.

Augmented Intelligence ↓

- It is the use of AI technologies to support & enhance human intelligence, rather than replace humans.
- It focuses on human + machine collaboration for better decision making.
- So together, they improve accuracy & efficiency.

Working ↓

- ① collects large amounts of data.
- ② Analyzes & finds patterns
- ③ Gives suggestions (or) predictions.
- ④ Human makes final decision.

Ex: - Medical diagnosis system suggesting diseases.

Role in CC

- It helps sys understand content & user needs.
- Improves human - computer interaction
- It acts as a bridge b/w human cognition & AI.

Advantages:

- Improves decision accuracy
- Increases productivity.
- Reduces human workload.
- Handles complex data easily.

Conclusion:

- Augmented intelligence is an imp concept in cognitive computing where humans & AI works together by combining both we get better results.

Cognition Modeling Paradigms:

- declarative / logic-based computational cognitive modeling.

⇒ Definition:

⇒ Among different paradigms, declarative (or) logic-based cognitive modeling focuses on

representing knowledge in a structured & logical form,
similar to human reasoning.

⇒ This approach represents knowledge using facts, rules & logical relationships.

- It is based on the idea that human cognition can be expressed using ~~logic~~ formal logic.

- In this the knowledge is stored in declarative form.

Working:-

- Knowledge^{is} stored as logical ~~structure~~ statements
- The sys applies rules to these statements.
- New conclusions are derived using inference.
- This mimics human logical reasoning.

Ex:-

- fact : Ram is a human
- Rule :- All human are mortal
- conclusion: Ram is mortal.

Techniques used:-

- Predicate logic
- Propositional logic
- Rule based & expert systems

Advantages:

- clear & structured representation of knowledge.
- Easy to understand & explain reasoning
- provides accurate logical conclusions.

Limitations:

- can't handle incomplete data effectively.
- Lacks flexibility compared to human thinking.
- Requires complete knowledge.

Applications:

- Expert sys
- Medical diagnosis sys
- Decision support sys.
- Knowledge based AI-sys.

Connectionist Models of cognition

→ These are also known as (or) called as neural network models.

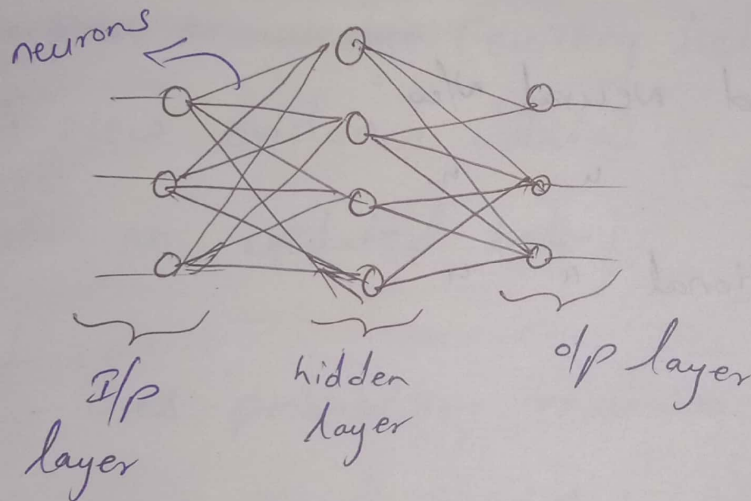
- These models try to simulate how the human brain works by using a n/w of simple interconnected units (like neurons).

→ In this approach knowledge is not stored as rules

(or) symbols.

- Instead, it is stored as patterns of activation across a net of nodes.

- The main idea is that learning happens by adjusting the connections (weights) b/w neurons based on experience.



⇒ It is inspired by ~~ANND~~ human brain & Based on ANN.

- It Learns from data using training methods like weight adjustment.
- It is Good at pattern recognition & generalization.

Working

- ① I/p is given to the N/w.
- ② It passes through multiple layers of neurons.
- ③ Each connection has a weight.
- ④ The sys -adjusts weights based on errors.
- ⑤ finally, it produces the correct o/p after learning.

Types

- feedforward neural N/w
- Recurrent " "
- Convolutional " "

Application

- speech recognition
- face " "
- handwriting " "
- language & memory modeling.

Bayesian Models of Cognition

⇒ It tells how humans think & make decisions under uncertainty using probability theory.

- These models are based on "Bayes's Theorem",
- It helps in updating beliefs when new info is received.

⇒ The main idea is that humans combine &

- prior knowledge (existing belief)
- New evidence (data).

to form an updated belief.

Characteristics

- uses probabilistic reasoning
- handles uncertainty & incomplete info.
- continuously updates beliefs with new data

Working

- Starts with an initial belief (prior)
- Receive new evidence
- Apply Bayes rule to update belief.
- Get a new probability (posterior)
- Repeat the process as more data comes.

Bayes theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

- $P(A|B)$ → probability of hypothesis 'A' given Evidence 'B'.
- $P(B|A)$ → likelihood of evidence given 'A'.
- $P(A)$ → prior probability of 'A'.
- $P(B)$ → total "u" of Evidence.

Applications

- Language understanding
- Decision making under uncertainty
- Visual perception.

A Dynamical Systems approach to cognition

→ It explains human thinking as a continuous, time-changing process rather than a step-by-step computation.

- It views cognition as the result of interaction b/w brain, body & environment.

⇒ Cognition is not fixed (or) static. It is ↓

- Dynamic
- Continuous.

- It focuses on time based changes in thinking.
- It uses mathematical equation to describe changes of cognitive

Working ↓

- The brain receives input from the environment
- Internal states continuously change over time.
- Small changes in input can lead to different o/p.
- Behavior "emerges" from this continuous interaction.

Ex:

- A person catching a ball:
 - They continuously adjust their movement based on the ball's position.
 - No fixed calculation needed.
 - The action is dynamic & real-time.

Applications ↓

- Robotics , - cognitive science research,
- Human motor behavior analysis
- Adaptive AI systems.